Math 121: Commutative Algebra

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1 Rings

Important Example: Let X be a "geometric set", and consider the ring

$$\mathrm{Hom}(X, \mathbb{R}) = \{X \xrightarrow{f} \mathbb{R}\}$$

with pointwise addition and multiplication

$$(f+g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x)$$

$r \in \mathrm{Hom}(X, \mathbb{R})$ is a unit if and only if there exists $g \in \mathrm{Hom}(X, \mathbb{R})$ such that $fg = 1$ if and only if $\forall x \in X, (fg)(x) = f(x)g(x) = 1$

$$\mathrm{Hom}(X, \mathbb{R})^\times = \mathrm{Hom}(X, \mathbb{R}^\times)$$

Definition 1.1. *Let* $a \in R$, $a \neq 0$

- a *is a unit if and only if* $\exists b \in R$ *such that* $ab = 1$,
- a *is a zero divisor if and only if* $\exists b \in R \setminus \{0\}$ *such that* $ab = 0$,

Lemma 1.2. *The set of units* $R^\times = \{a \in R \,|\, a$ *is a unit*$\}$ *is a group under multiplication.*

Proof: Worksheet (03/31).

Lemma 1.3 (Useful Lemma)**.** *If* $\phi : R \to S$ *is a ring homomorphism, then it follows from the definition that*

- $\phi(0_R) = 0_S$
- $\phi(-a) = -\phi(a)$
- $\phi(a^{-1}) = \phi(a)^{-1}$

Remark 1.4. $\mathbb{Z}$ *is initial in the category of rings. There is a unique homomorphism from the initial object to every other object. For any ring* R,

$$\mathsf{Hom}(\mathbb{Z}, R) = \{1 \to 1_R\}$$

We don't know anything about homomorphisms to the initial obejct.

2 Ideals

Lemma 2.1. *If* $\phi : R \to S$ *is any ring homomorphism and* $J \subset S$ *is an ideal* $\phi^{-1}(J)$ *is also an ideal.*

Proof: Worksheet Q4.1.

Theorem 2.2. *If* $I \subset R$ *is an ideal then,*

1. *There exists a ring* R/I *and a surjective ring homomorphism* $\pi : R \to R/I$ *such that* $\ker(\pi) = I$. *Moreover* R/I *and* π *are unique upto isomorphism.*

2. *There is an inclusion preserving bijection*

$$\{\textit{Ideals in } R/I\} \leftrightarrow \{\textit{Ideals in } R \textit{ containing } I\}$$

$$\bar{J} \leftrightarrow \pi^{-1}(\bar{J})$$

Proof.

1. Construct R/I explicitly.

2. I gives an equivalence relation $\sim$ on R such that

$$a \sim b \iff a - b \in I \iff a = b + I$$

Show that equivalence classes $R/\sim$ is a ring.

$$R \to R/I = R/\sim$$
$$a \mapsto \bar{a} = a + I$$

Let $\bar{J}$ be an ideal in R/I, and J be an ideal in R and define the maps

$$\phi : \bar{J} \to \pi^{-1}(\bar{J}), \quad \psi : J \to \pi(J)$$

Now we want to show that $\phi \circ \psi = \psi \circ \phi = \mathrm{id}$. Consider $\phi \circ \psi(J)$ and $\psi \circ \phi(\bar{J})$. Note that

$$\psi \circ \phi(\bar{J}) = \pi(\pi^{-1}(\bar{J})) = \bar{J}$$

since π is surjective. Thus $\psi \circ \phi = \mathrm{id}$.

Now, note that $J \subseteq \phi \circ \psi(J) = \pi^{-1}(\pi(J))$, and since $\bar{0} \in \pi(J)$, $I \subseteq \pi^{-1}(\pi(J))$. An element $x \in \pi^{-1}(\bar{J})$ looks like $x + I$, and $\pi^{-1}(\bar{J}) = J + I$ Moreover since $\bar{0} \in \bar{J}$, we have $\pi^{-1}(\bar{0}) \subseteq \pi^{-1}(\bar{J})$ which implies $I \subseteq J$. Thus, $\psi \circ \phi(J) = J$ and $\psi \circ \phi = \mathrm{id}$.

$\square$

Exercise: Describe the ideals in

1. $\mathbb{Z}/\langle 24\rangle = \{\langle 0\rangle, \langle 1\rangle, \langle 2\rangle, \langle 3\rangle, \langle 4\rangle, \langle 6\rangle, \langle 8\rangle, \langle 12\rangle\}$
2. $\mathbb{Z}/\langle n\rangle = \{\langle d\rangle : d|n\}$
3. $\mathbb{Z}/\langle 3\rangle = \{\langle 0\rangle, \langle 1\rangle\}$
4. $\mathbb{Z}/\langle 5\rangle = \{\langle 0\rangle, \langle 1\rangle\}$
5. $\mathbb{C}[x,y]/\langle y - x^2\rangle \cong \mathbb{C}[x] = \{\text{Ideals in } \mathbb{C}[x]\}$
6. $\mathbb{C}[x]/\langle x^n\rangle = \{\langle x^i\rangle : 0 \leq i < n\}$

Lemma 2.3. *If $\phi : R \to S$ is a ring homomorphism with $\ker(\phi) \supseteq I$ then there exists a unique morphism $\widetilde{\phi} : R/I \to S$*

$$\begin{tikzcd} R \arrow[r, "\pi"] \arrow[d, "\phi"'] & R/I \arrow[dl, dashed, "\widetilde{\phi}"] \\ S & \end{tikzcd}$$

Universal Property of Quotinents: *Upto isomorphism R/I is the ring such that*

$$\mathrm{Hom}(R/I, S) = \{\phi : R \to S : I \subset \ker(\phi)\}$$

We want to define $\widetilde{\phi}(\overline{a}) = \widetilde{\phi}(a + I) = \phi(a)$ and $\widetilde{\phi}(\overline{0}) = 0$.

Lemmact 2.4. *Maximal Ideal $\implies$ Prime Ideals.*

Definition 2.5. *The spectrum of a ring R is the set*

$$\mathrm{Spec}(R) = \{\textit{Prime Ideals in } R\}$$

the maximal spectrum of R is the set

$$\mathfrak{m}\,\mathrm{Spec}(R) = \{\textit{Maximal Ideals in } R\}$$

$\mathrm{Spec}(R)$ *is where the 'geometry lives'. Note that $\mathfrak{m}\,\mathrm{Spec}(R) \subset \mathrm{Spec}(R)$.*

Example: $R = \mathbb{Z}[\sqrt{-5}]$. 2 *is irreducible but not prime.* $R/\langle 2\rangle = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Example: $R = \mathbb{C}[x,y]/\langle x^3 - y^2\rangle$. x *is irreducible, but $\langle x\rangle$ is not prime.* $R/\langle x\rangle = \mathbb{C}[y]/\langle y\rangle$ *and y is a zero divisor.*

Theorem 2.6. *Let R be a non-zero ring.*

1. *R has a maximal ideal.*
2. *If $I \subsetneq R$ is a proper ideal there exists a prime ideal $P \subseteq R$ such that $I \subset P \subsetneq R$.*

Proof. **Zorn's Lemma:** Let $(P, \leq)$ be a po-set.

If $P \neq \emptyset$ and if $C = C_1 \leq C_2 \leq C_3 \leq \cdots$ is a chain in $(P, \leq)$ then there exists $B \in P$ such that $C_i \leq B$ for all i, then there exists $M \in P$ such that if $S \leq M$ then $M = S$.

Let $P = \{$Proper ideals in $R\}$ and $\leq$ be inclusion. We claim that $(P, \leq)$ is a poset.

1. - $P \neq \emptyset$ since $\langle 0 \rangle \in P$.
   - Given a chain in P

     $$I_1 \subseteq I_2 \subseteq \cdots$$

     Then, define $B = \bigcup I_j$. Note that B is an ideals since union of nested ideals is an ideal. And B is proper since $1 \notin I_j$ for all j, and hence $1 \notin B$. Then there exists some ideal $M \subset R$ such that M is maximal.
2. Follows from (1). Let $R = R/I$. Pre-image of prime ideals is prime.

Thus, $\mathfrak{m}\,\mathrm{Spec}(R) \neq \emptyset$ and $\mathrm{Spec}(R) \neq \emptyset$. □

3 Zariski Topology

Definition 3.1. *If $S \subseteq R$ is any subset of R,*

$$\mathbb{V}(S) = \{[P] \in \mathrm{Spec}(R) | S \subseteq P\} \subseteq \mathrm{Spec}(R)$$

Definition 3.2. *An algebraic subset of $\mathbb{K}^n$ of some field K is a set of the form*

$$\mathbb{V}(\mathcal{S}) = \{P \in \mathbb{K}^n | f(p) = 0, \forall f \in \mathcal{S}\}$$

for a subset $\mathcal{S} \subset \mathbb{K}[x_1, \cdots, x_n]$

Examples:

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^2$

$$\{(x, y) \in \mathbb{R}^2 | x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2+y^2-1) \subseteq \mathbb{R}^3$

$$\{(x,y,z) \in \mathbb{R}^3 | x^2+y^2-1=0\}$$

- $\mathbb{V}(x^2-y) \subseteq \mathbb{R}[x,y]$

$$\{(x,y) \in \mathbb{R}[x,y] | x^2=y\}$$

- $\mathbb{V}(x^2-xy) \subseteq \mathbb{R}[x,y]$

$$\{(x,y) \in \mathbb{R}[x,y] \,|\, x=0,\ x-y=0\}$$

- $\mathbb{V}(x^2+1) \subseteq \mathbb{R}[x]$

$$\mathbb{V}(x^2+1)=\emptyset$$

- $\mathbb{V}(x^2+1) \subseteq \mathbb{C}[x]$

$$\mathbb{V}(x^2+1)=\{i,-i\}$$

- $\mathbb{V}(x^2+4) \subseteq \mathbb{F}_3[x]$

$$\mathbb{V}(x^2+4)=\emptyset$$

- $\mathbb{V}(x^2+4) \subseteq \mathbb{F}_p[x]$
  TODO:
- $\mathbb{V}(x^6-3x^5y+3x^4y^2-x^3y^3) \subseteq \mathbb{R}[x,y]$

$$\mathbb{V}(x^6-3x^5y+3x^4y^2-x^3y^3)=\mathbb{V}(x^3(x-y)^3)=\{(x,y) \in \mathbb{R}[x,y] \,|\, x=0,\ x-y=0\}$$

- $\mathbb{V}(n(x^2+y^2+z^2), n \in \mathbb{Z}) \subseteq \mathbb{R}^3$
  TODO:
- $\mathbb{V}(\text{a collection of any linear polynomials in } \mathbb{K}[x_1,\cdots,x_n])$
  TODO:

Definition 3.3. *Let $V \subseteq \mathbb{K}^n$ be an algebraic set. The defining ideal of V is*

$$\mathbb{I}(V)=\{f \in \mathbb{K}[x_1,\cdots,x_n] \,|\, f(p)=0, \forall p \in V\}$$

Definition 3.4. *If* $\mathfrak{M} \subseteq \mathsf{R}$ *is a maximial ideal, the residue field of* $\mathfrak{M}$ *is* $\mathsf{R}/\mathfrak{M}$.

Definition 3.5. *Let* $\mathbb{K}$ *be a field. A* $\mathbb{K}$ *algebra is the data of a ring* R *together with a fixed ring homomorphism*

$$\alpha : \mathbb{K} \to \mathsf{R}$$

Definition 3.6. *If* $(\mathsf{R}, \mathbb{K} \xrightarrow{\alpha} \mathsf{R})$ *and* $(\mathsf{S}, \mathbb{K} \xrightarrow{\beta} \mathsf{S})$ *are* $\mathbb{K}$*-algebras, a ring map* $\phi : \mathsf{R} \to \mathsf{S}$ *is a morphism of* $\mathbb{K}$*-algebras iff*

$$\begin{tikzcd} \mathsf{R} \arrow[r, "\phi"] & \mathsf{S} \\ \mathbb{K} \arrow[u, "\alpha"] \arrow[ur, "\beta"] & \end{tikzcd}$$